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4 August 1971
SUPERSEDING
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25 November 1958

MILITARY SPECIFICATION

SEALING COMPOUND, JOINT, TWO COMPONENT, FOR BOLTED ALUMINUM OR STEEL PETROLEUM-STORAGE TANKS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a sealing compound for sealing the seams of bolted aluminum or steel petroleum-storage tanks.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

H-B-420	- Brush, Paint, Flat, Metal-Bound.
QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.

Military

MIL-T-3689	- Tubes, Collapsible.
MIL-T-10086	- Tanks, Liquid Storage, Metal; Vertical, Bolted.

FSC 8030

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STANDARDS

Federal

FED. TEST METHOD STD. No. 601 - Rubber: Sampling and Testing.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D-381 - Test for Existent Gum by Jet Evaporation.
D-1084 - Consistency of Adhesives, Tests for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

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3. REQUIREMENTS

3.1 Description. The sealing compound shall be a two component system to be used to seal seams in storage tanks specified in MIL-T-10086. The curing component (A) shall be made up of a curing agent such as a powder dispersed in an extractable plasticizer, in creme form in strength and volume to fully cure a unit of base component (B). The base component B shall be a stable liquid polymer placed in an oxygen-displaced, airtight container which will have a shelf-life of 1 year when stored at temperatures under 100° F. as specified herein. The base component container shall provide space to accomodate the component A, curing agent, and allow mixing in the container by simple mechanical or manual stirring techniques to form a homogeneous sealing compound. Skinning of the base component in the container shall be not greater than 1/8 of an inch in thickness.

3.2 Qualification. Sealing compound furnished under this specification shall be products which have been tested, have passed the qualification tests specified herein, and have been listed on or approved for listing on the applicable qualified products list (see 6.5 and 6.6).

3.2.1 Accelerated life. For qualification, the components of the sealing compound in the supplier's oxygen displaced airtight containers shall be conditioned as specified in 4.5.2.1 to assure stability in shipping and storage as specified herein (see 3.1).

3.3 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring activity (see 6.4).

3.4 Brushing properties. When tested as specified in 4.5.3.1, the sealing compound shall flow freely in brushing to form an even, uninterrupted surface for not less than 60 minutes after mixing as specified in supplier's instructions.

3.5 Viscosity. When tested as specified in 4.5.3.2, the mixed sealing compound shall have a viscosity within the range of 40,000 to 80,000 centipoises at laboratory atmospheric temperature as specified herein.

3.6 Cured compound.

3.6.1 Resistance to water, aromatic fuels, and vapors. When tested as specified in 4.5.3.3, the sealing compound shall show no evidence of softening, blistering, or separation from the panel, or evidence of corrosion of the metal under the coating in the test specimens.

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3.6.2 Low temperature flexibility. When tested as specified in 4.5.3.4, the cured sealing compound shall not crack, check, or lose adhesion in flexure.

3.6.3 Leak. When tested as specified in 4.5.3.5, the sealing compound shall show no evidence of leaking.

3.6.4 Hardness. When tested as specified in 4.5.3.6, the cured sealing compound shall have a minimum durometer hardness of 25.

3.6.5 Aged hardness. When tested as specified in 4.5.3.7, the cured sealing compound shall have a durometer hardness of not greater than 70.

3.6.6 Adhesion. When tested as specified in 4.5.3.8, the cured sealing compound shall have an adhesion value of not less than 5 pounds per inch of width when stripped from steel and aluminum panels.

3.6.7 Weight retention. When tested as specified in 4.5.3.9, the compound shall retain not less than 90 percent of the original weight of the specimens.

3.6.8 Volume change in aromatic fluid. When tested as specified in 4.5.3.10, the sealing compound shall have a change in volume not greater than 5 percent of the original volume of the specimens.

3.6.9 Extraction in aromatic fuel. When tested as specified in 4.5.3.11, the cured sealing compound shall have an extractable content not greater than 5 percent by weight.

3.6.10 Fuel contamination. When tested as specified in 4.5.3.12, the nonvolatile residue after evaporation, shall be not greater than 5 mgs per 100 mls of test medium No. 6. The stove evaporated residue shall be not greater than 2 mgs per 100 mls of test medium No. 6.

3.6.11 Nonvolatile content. The combined nonvolatile content of both components of the sealing compound, when tested separately as specified in 4.5.3.13, shall be not less than 95 percent of the total weight of the sealing compound mixed in ratio as recommended by the supplier.

3.6.12 Mixing and storage instructions. Each container of component A and B shall be permanently and legibly marked with the suppliers recommended instructions for mixing. A warning shall be printed legibly on each container stating that the compound shall be stored under 100° F. and that mixed sealing compound shall be used within 60 minutes of preparation. Labeling shall be by stencil, lithographing, or other permanent method. All containers of sealing compound components shall have the following warning: "DO NOT USE IN CONTACT WITH POTABLE WATER."

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Qualification inspection.

4.3.1 Sampling. The supplier shall submit a minimum of six quart cans of base component B and six unit containers of curing agent component A (see 3.2). All containers shall be identified and marked as follows:

Manufacturers name
Date of manufacture
Batch number
Sequence of sampling numbers
Instructions for mixing.

4.3.2 Examination. Samples specified in 4.3.1 shall be examined for stability of components in the oxygen-displaced air-tight containers. Hardening of resins that will not allow manual mixing, or a skinning over the surfaces greater than 1/8 of an inch, shall be cause for rejection.

4.3.3 Tests. Samples specified in 4.3.1 shall be tested as specified in 4.5.3.1 through 4.5.3.13. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Batch. A batch shall consist of all the sealing compound formulated in a single or unit mix of base material component B, and in a single or unit mix of curing agent component A, and offered for delivery by the supplier.

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4.4.2 Sampling. Two 1-quart samples of base component B and two unit containers of curing component A shall be selected at random from each batch of sealing compound for examination and testing.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection of the batch of sealing compound from which the samples were taken.

4.4.4 Tests. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.3.2, 4.5.3.3, 4.5.3.8, 4.5.3.11 and 4.5.3.12. Failure of any test by either sample shall be cause for rejection of the batch of sealing compound from which the samples were taken.

4.5 Inspection procedure.

4.5.1 Examination. The sealing compound shall be examined for the following defects:

101. Mixing ratio not as specified.
102. Skinning of resins greater than 1/8 inch.
103. Resins will not mix homogeneously.
104. Labeling not as specified.

4.5.2 Test conditions and specimens.

4.5.2.1 Accelerated life. For qualification testing (see 3.2), samples of the sealing compound shall be conditioned for not less than 30 days at 120° F. plus or minus 2° F., and then conditioned for a minimum of 24 hours at laboratory atmospheric conditions specified in 4.5.2.2. Conditioned samples shall be used in performing all tests specified in 4.5.3.1 through 4.5.3.13.

4.5.2.2 Laboratory atmospheric conditions. Unless otherwise specified herein, tests shall be run at an atmospheric temperature of 73.5° F. plus or minus 2° F. and 50 percent plus or minus 2 percent relative humidity.

4.5.2.3 Mixing. The two components of the sealing compound shall be combined in accordance with instructions marked on the container. The two components shall be stirred thoroughly by hand for 5 minutes or until no specks or streaks of unmixed materials are visible but in no case for more than 10 minutes.

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4.5.2.4 Curing. Test specimens shall be cured for 70 hours at laboratory atmospheric conditions (see 4.5.2.2). Tests shall be initiated within 2 hours after curing of the test specimens.

4.5.2.5 Panels,disks and rings. Steel for panels and disks shall be cold-rolled, mild-carbon steel, and shall conform to QQ-S-698 with a bright finish. Aluminum Alloy for panels and rings shall conform to QQ-A-250/2, temper F. Before the panels, disks, and rings are used in the test determinations, they shall be free of scale, rust, and other imperfections, shall be polished with 2/0 abrasive garnet paper, and cleaned of oil, grease, and dirt with toluene.

4.5.2.6 Sealing compound strips. Sealing compound mixed as specified in 4.5.2.3 shall be spread onto a lightly lubricated steel plate to form a 50 square-inch sheet. After curing as specified in 4.5.2.4 and buffing, if necessary, the sheet of sealing compounds shall be smooth and have a thickness of 0.050 to 0.065 inch. The cured sheet shall be stripped from the plate and cut into 1- by 2-inch strips.

4.5.3 Test procedures.

4.5.3.1 Brushing properties. Three specimens from the sets of sample components A and B of a batch of sealing compound shall be used for this test. Each specimen shall be of sufficient quantity to make 12 fluid ounces of sealing compound. One specimen of each component shall be conditioned for 16 hours at plus 20° F. plus or minus 2° F.; one specimen of each component for 16 hours at laboratory atmospheric conditions specified in 4.5.2.2, and one specimen of each component for 16 hours at plus 95° F. plus or minus 2° F. A steel panel shall be conditioned with the specimens. While still at the conditioning temperature, the respective conditioned specimens shall be mixed as specified in 4.5.2.3, allowed to stand undisturbed for 1 hour, stirred for 1 minute, and then applied to the steel panel using a 3-inch brush conforming to H-B-420, grade C. Nonconformance to 3.4 shall constitute failure of this test.

4.5.3.2 Viscosity. The viscosity of the sealing compound shall be determined with a Brookfield viscometer, model LVF, run at 6 rpm and using a No. 4 spindle or equivalent apparatus, and a container having a diameter of not less than 2-1/4-inches. The procedure shall be as follows:

(a) Prepare a specimen consisting of not less than 1 pint of sealing compound as specified in 4.4.2.3. Nine minutes after the start of mixing, stir vigorously by hand for 1 minute.

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(b) Mount the viscometer vertically so that the spindle is immersed in sealing compound to the groove cut in the shaft.
(c) Start the motor exactly 5 minutes after the final mixing (15 minutes after combining the two components).
(d) The first viscosity reading shall be taken exactly 1 minute after starting the motor, and the second and third readings shall be taken on the following two successive revolutions of the dial.

Reading may be taken on either the 100 scale or the 500 scale. If the 100 scale is used, multiply the reading by 1,000 to obtain the viscosity in centipoises. If the 500 scale is used, multiply the reading by 200 to obtain the viscosity in centipoises. The viscosity shall be the average of the three readings. Nonconformance to 3.5 shall constitute failure of this test.

4.5.3.3 Resistance to water, aromatic fuels, and vapors. Two 1-inch by 6-inch steel panels and two 1-inch by 6-inch aluminum panels as specified in 4.5.2.5 shall be coated on one side with sealing compound prepared as specified in 4.5.2.3. After curing as specified in 4.5.2.4, the sealing compound shall be 0.030 to 0.040 inch thick. The coated panel shall be placed in a covered glass vessel containing a layer of tap water and a layer of test medium No. 6 conforming to FED. TEST METHOD STD. No. 601, method 6001. The panels shall be placed in the vessels in such a manner that 2 inches of each panel shall be immersed in water, 2 inches in test medium No. 6, and 2 inches exposed to air-vapor mixture. The panels shall be exposed to these conditions for not less than 14 days with the temperature maintained at plus 140° F. plus or minus 2° F. The coatings shall then be examined for softening, blistering, and separation from the panel. The coating compound shall be stripped from the panels using cyclohexanone or trichloroethylene, and the panels shall be examined for visual evidence of corrosion. Two determinations shall be made. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.3.4 Low temperature flexibility. Three 1/32-inch by 1-inch by 6-inch aluminum panels shall be prepared as specified in 4.5.2.5 and coated on one side with sealing compound mixed as specified in 4.5.2.3. After curing as specified in 4.5.2.4 the coatings shall be 0.050 to 0.065 inch thick. The panels shall be immersed for 46 hours in test medium No. 6 conforming to FED. TEST METHOD STD. No. 601, method 6001, at plus 120° F. plus or minus 2° F. The panels shall then be dried in laboratory air for 4 hours, dried in a circulating air oven at plus 100° F. plus or minus 2° F. for 18 hours, and then conditioned at minus 65° F. plus or minus 2° F. for 4 hours; a 4-inch diameter (nominal size) mandrel shall be conditioned with the test panels. Within 1 minute after removal from low temperature conditions, the panels shall be bent 180 degrees around the mandrel with the coated side outward. The coatings shall then be examined, and the results shall be compared. Nonconformance to 3.6.2 shall constitute failure of this test.

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4.5.3.5 Leakage. The sealing compound shall be applied to two pairs of disks as shown on figure 1 and cured as specified in 4.5.2.4. One pair of disks shall be immersed for 30 days in water, and the other pair shall be immersed for 30 days in test medium No. 6. The temperature of the water and the test medium shall be maintained at plus 73.5° F. plus or minus 2° F. At the end of the immersion period, each pair of disks shall be attached to the metal cup (see figure 1). The cup shall be filled with test medium No. 6. The pressure in the cup shall be slowly increased to 15 psig and shall be maintained at this pressure for 22 to 24 hours. At the end of the time period, the apparatus shall be examined for leakage of fluid. Nonconformance to 3.6.3 shall constitute failure of this test.

4.5.3.6 Hardness. Sealing compound strips, prepared as specified in 4.5.2.6, shall be buffed to a thickness of 0.250 to 0.300 inch and allowed to set for 1 hour at laboratory atmospheric conditions specified in 4.5.2.2. The hardness of the buffed specimens shall be determined in accordance with FED. TEST METHOD STD. No. 601, method 3021. Nonconformance to 3.6.4 shall constitute failure of this test.

4.5.3.7 Change in hardness. A specimen shall be prepared by using strips made as specified in 4.5.2.6 and cured as specified in 4.5.2.4. The cured specimens shall be aged 22 to 24 hours at 180° F. plus or minus 5° F. The strips shall be allowed to set for 1 hour at laboratory atmospheric conditions specified in 4.5.2.2. The strips after setting shall be buffed to a thickness of 0.250 to 0.300 inch, and the hardness shall be determined as specified in FED. TEST METHOD STD. No. 601, method 3021. Nonconformance to 3.6.5 shall constitute failure of this test.

4.5.3.8 Adhesion. Three steel and three aluminum panels, each 3 inches by 6 inches by 1/16 inch, shall be polished and cleaned on one side as specified in 4.5.2.5. Six 3-inch by 12-inch strips of cotton cloth shall be prepared. A coating of sealing compound mixed in accordance with 4.5.2.3 shall be applied 0.050 to 0.065 inch thick to the polished and cleaned face of each metal panel and to half of one face of each cotton strip. The cotton strip shall be placed, coated side downward, onto the coated face of the metal strip leaving a loose end 6 inches long. The sealing compound shall be worked into the cotton strips by rolling lightly with a hand roller. The coating shall be cured as specified in 4.5.2.4. After curing, the coating of sealing compound and cotton cloth shall be cut into three 1-inch strips running parallel to the 6-inch edge of the panel. The two outer strips shall be removed leaving a 1-inch strip in the center of the panel. The prepared specimen shall be immersed at plus 73.5° F. plus or minus 2° F. for 14 days in 1 quart of tap water and then immersed at plus 73.5° F. plus or minus 2° F. for 14 days in 1 quart of test medium No. 6 conforming to FED. TEST METHOD STD. No. 601, method 6001. While the specimen

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is still wet, the loose end of the cotton strip shall be folded back on the part bonded to the plate. Then the cotton strip with sealing compound shall be separated from the metal panel for 1 inch at the folded end. The loose end of the cotton strip of cloth shall be clamped in one grip of the testing machine specified in method 8011 of FED. TEST METHOD STD. No. 601, and the adjacent end of the metal panel shall be clamped in the other grip. The adhesion shall be the force in pounds per inch of width required to strip the coating from the metal panel at an angle of 180 degrees. The adhesion of each specimen shall be determined within 5 minutes after removal from the test medium. The average of five peak loads shall be considered the adhesion value for a single specimen. If failure occurs in the coating or at the fabric-coating interface, the test machine shall be stopped. The coating shall be cut with a knife through to the metal-coating interface along the line of separation, and the test machine shall be started again. If failures of this type occur repeatedly, the adhesion of the specimen shall be recorded as greater than the average of the peak loads so obtained. The average of the adhesion values obtained on three aluminum panels and the average of the adhesion values obtained on three steel panels shall be compared with 3.6.6. Nonconformance to 3.6.6 shall constitute failure of this test.

4.5.3.9 Weight retention. Three sealing compound strips, prepared as specified in 4.5.2.6, shall be weighed to the nearest milligram to determine (W_1). The strips shall be aged for 22 to 24 hours at a temperature of plus 180° F. plus or minus 2° F. in accordance with FED. TEST METHOD STD. No. 601, method 7221. After aging, the strips shall be allowed to set for 1 hour at laboratory atmospheric conditions as specified in 4.5.2.2. The aged strips shall be weighed to the nearest milligram to determine (W_2). The weight retention shall be calculated as follows:

$$\text{Percent weight retained} = \frac{W_2 \times 100}{W_1}$$

The percent weight retained shall be averaged, and the average obtained shall be compared with 3.6.7. Nonconformance to 3.6.7 shall constitute failure of this test.

4.5.3.10 Volume change in test medium No. 6. The volume change shall be determined in accordance with FED. TEST METHOD STD. No. 601, method 6211 using test medium No. 6 of method 6001 on strips of sealing compound conforming to 4.5.2.6. Three strips shall be immersed at a temperature of plus 73.5° F. plus or minus 2° F. for 22 hours plus or minus 1/4 hour. Nonconformance to 3.6.8 shall constitute failure of this test.

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4.5.3.11 Extraction in test medium No. 6. Sealing compound strips shall be made in accordance with 4.5.2.6. After curing as specified in 4.5.2.4, the sealing compound shall be removed from the plate. The thickness of the cured sheet shall be 0.030 to 0.040 inch and shall be as uniform as possible. Buffing the specimens to the required thickness will be permitted. Three 2-inch by 2-inch specimens shall be cut from the cured sheets and weighed to the nearest milligram. Each specimen shall then be wrapped in a 4-inch by 5-inch sheet of No. 1 filter paper without folding the specimen. Each wrapped specimen shall be rolled loosely and inserted in a separate Soxhlet extractor. The extractor shall have a 40-mm internal diameter, a 205-mm body length, and shall be fitted with an Allihn condenser and a 500-ml round-bottom flask. The standard-type joint at the top of the apparatus shall be 45/50, and the joint at the bottom shall be 24/40. The extractor shall be emptied once every 3 to 4 minutes. The specimen shall be extracted continuously for 22 to 24 hours by test medium No. 6. After extraction for 22 to 24 hours, the specimens shall be removed from the apparatus and wrapped in filter paper, air dried to constant weight (a period of not less than 46 hours) at laboratory atmospheric conditions specified in 4.5.2.2, and then weighed to the nearest milligram. The percent extraction shall be calculated as follows:

$$\text{Percent extraction} = \frac{(A-B)\ 100}{(A)}$$

Where:

A = original weight.
B = weight after extraction and drying.

Compare the average percent extraction of the three specimens with 3.6.9. Nonconformance to 3.6.9 shall constitute failure of this test.

4.5.3.12 Fuel contamination. Two 1-inch by 5-inch steel panels shall be prepared as specified in 4.5.2.5 and coated with sealing compound mixed as specified in 4.5.2.3. After curing as specified in 4.5.2.4, the coatings shall be buffed to 0.050 to 0.065 inch thick. The two panels shall be immersed for 46 hours in a single vessel containing 250 ml of test medium No. 6 conforming to FED. TEST METHOD STD. No. 601, method 6001. The temperature of the fluid shall be maintained at plus 73.5° F. plus or minus 2° F. The fluid shall then be decanted, and the amount of nonvolatile residue (existent gum) shall be determined in accordance with ASTM D381, except that total evaporation time shall be 45 minutes and temperature shall be plus 320° F. to plus 329° F. The beaker containing the nonvolatile residue shall then be placed for 30 minutes in a bath maintained at 572° F. plus or minus 10° F. After

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the nonvolatile residue is cooled to laboratory atmospheric conditions (see 4.5.2.2) in a closed container, the beaker shall be weighed and the stove gum residue shall be calculated for 100 ml of contaminated fluid. Nonconformance to 3.6.10 shall constitute failure of this test.

4.5.3.13 Nonvolatile content. The nonvolatile content (total solids content) of the base component and the curing component shall be determined in accordance with ASTM D-1084, except that the nonvolatile content of the base component and the curing component shall be determined separately. The combined nonvolatile content shall be calculated, as follows:
The nonvolatile content of the sealing compound in percent

$$(N_a)\ (W_a) + (N_b)\ (W_b)$$

Where

N_a = nonvolatile content of the curing component in percent.
W_a = weight fraction of the curing component in the sealing compound.
N_b = nonvolatile content of the base component in percent.
W_b = weight fraction of the base component in the sealing compound.

Nonconformance to 3.6.11 shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105.	Materials and containers not as specified.	Each incorrect material or container shall be considered one defect.
106.	Component B, not packaged in quantity as specified herein, and component A not packaged in quantities recommended by the specific supplier.	
107.	Tubes of curing component A not secured to quart container as specified.	
108.	Intermediate packages not waterproof sealed as specified for level A.	
109.	Strapping not zinc coated for level A packing.	
110.	Weight limitation of the containers exceeded for packing.	
111.	Marking illegible, incomplete, or incorrect.	

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified.

5.1.1 Level A. The sealing compound shall be packaged in quantities such that the combined volume of component B and the suppliers recommended volume of component A shall not exceed 32 fluid ounces. Component B shall be packaged as specified in quart cans conforming to PPP-C-96, type V, class 2. The curing agent or component A shall be packaged in quantities of 6 fluid ounces or less as specified herein in tubes conforming to MIL-T-3689, class 4 or 5. Components A and B shall be unitized by having a tube of the curing agent (component A) secured to the quart can of component B in the vertical position with one 6-inch (minimum) length of tape of minimum 2-inch width conforming to PPP-T-60, type III or IV, class 1. The unitized sealing compound packaged as specified shall be intermediate packaged in a box conforming to PPP-B-636, V3c. The basic packaging unit shall be of a size to accomodate 12 quart cans of component B as specified herein. The plastic tube of curing agent (component A) taped to the can of component B shall be rotated to place the tube in the void between two cans of component B to prevent free movement and crushing in shipping and handling. The boxes shall be closed and waterproof sealed with tape as specified for slotted boxes in accordance with the appendix to the box specification.

5.1.2 Level C. The sealing compound, and curing agent shall be packaged to afford protection against damage and deterioration from the supplier to the first receiving activity. The supplier may use his standard practice when it meets these requirements. The sealing compound shall be packaged in 1-quart cans, and the curing agent shall be packaged in quantities of 6 fluid ounces or less in collapsible tubes. One can of sealing compound and one tube of curing agent, when applicable, shall be unitized by securing the curing agent tube in a vertical position to the side of a 1-quart can with tape. The sealing compound unitized with the curing agent, shall be placed, in quantities of 12 , in a close-fitting fiberboard box having a minimum bursting strength of 200 psi.

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. The sealing compound packaged and intermediate packaged as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style I, in quantities not to exceed the weight limitation of the container. Strapping shall be zinc coated.

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5.2.2 Level B. The intermediate container (see 5.1) used to package the sealing compound shall serve as the shipping container. Strapping need not be zinc coated. When specified, the intermediate containers shall be consolidated in a box conforming to PPP-B-601, class domestic, or PPP-B-640, class 2.

5.2.3 Level C. The sealing compound, and the curing agent, packaged as specified, shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules. When specified, the intermediate container (see 5.1) used to package the sealing compound, and the curing agent, shall serve as the shipping container. At the option of the supplier, the intermediate package may be consolidated in containers to assure acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or purchase order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The sealing compound covered by this specification is intended for sealing the bottom seams of bolted aluminum or steel tanks used for storing gasoline, jet fuel and diesel fuel oil only.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) When intermediate packaging will be consolidated for level B packing (see 5.2.2 and 5.2.3).
(d) Any special marking required (see 5.3).

6.3 Level C packaging and packing are not be be used for Army procurement unless it is known in advance that such procurement will be solely for CONUS use.

6.4 The contracting officer should require suppliers to furnish certification or evidence that the compound conforms to the toxicity requirements of 3.3.

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6.5 Qualification.

6.5.1 When the sealing compound covered by this specification is procured as an end item by the Government, only bids or proposals offering sealing compounds which are qualified under this specification (not necessarily listed on the QPL) at the time set for opening of the bids or award of a negotiated contract should be considered in making award.

6.5.2 When the sealing compound covered by this specification is a component of an end item being procured by the Government, the requirements of ASPR 1-1107.2(b) shall apply.

6.5.3 The attention of suppliers is called to the above conditions and manufacturers are urged to arrange to have the sealing compound that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for sealing compounds covered by this specification. The activity responsible for the Qualified Products List for this specification is: Commanding General, U. S. Army Mobility Equipment Command, Directorate of Research, Development and Engineering, Fort Belvoir, Virginia 22060. Attention: AMSME-RZM.

6.6 Retest and Re-examination. Determination has been made that all products listed on QPL 14231-2 dated 18 August 1959 no longer comply with the requirements of this revision, and that requalification is required.

Custodians:
Army - ME
Navy - YD
Air Force - 84

Review activity:
Army - MR

User activity:
Navy - SH

Preparing activity:
Army - ME

Project No. 8030-0336

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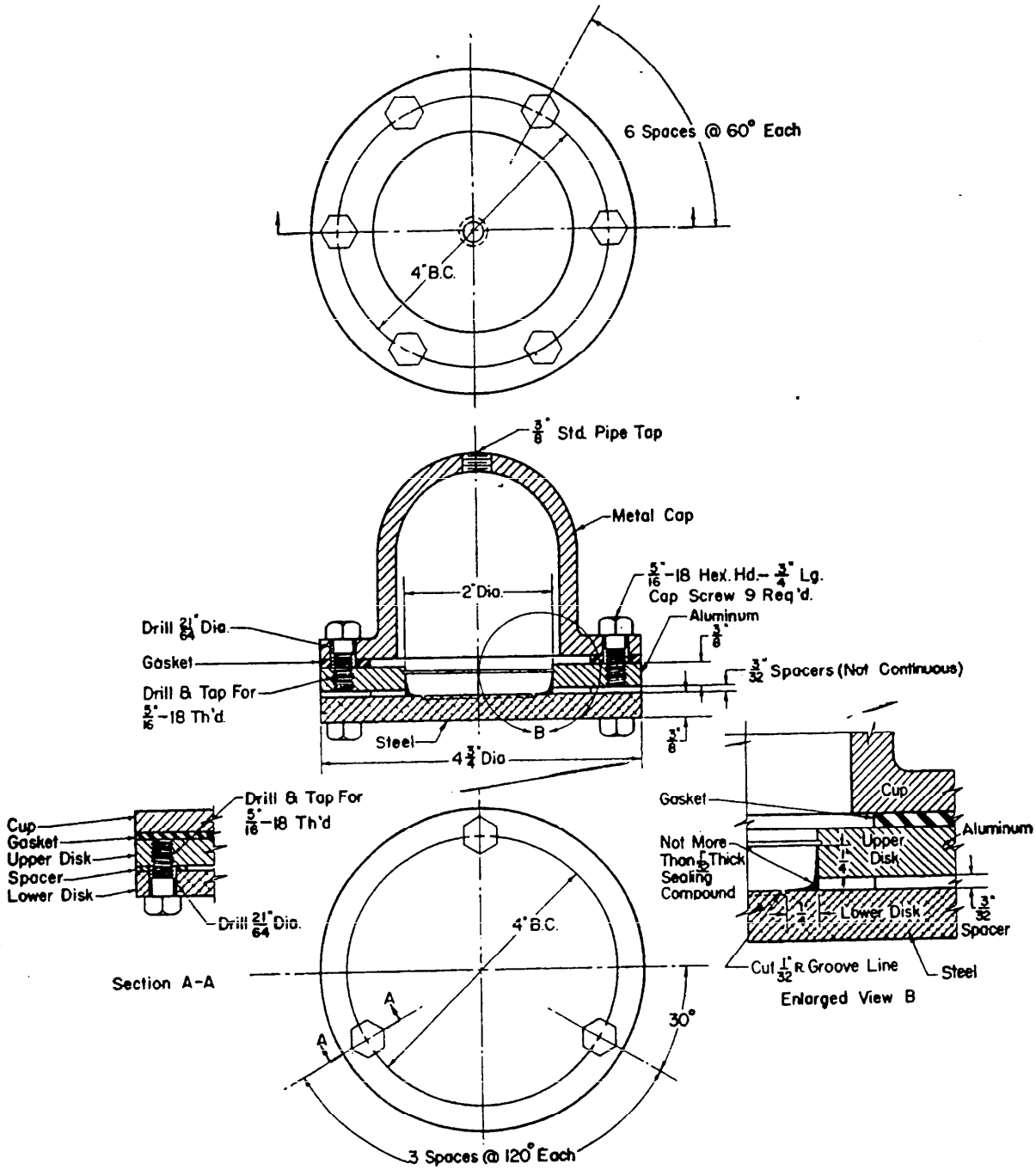


FIGURE 1. *Apparatus for leak test.*

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-14231C Sealing Compound, Joint, Two Component, For Bolted Aluminum or Steel Petroleum-Storage Tanks

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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DEPARTMENT OF THE ARMY
U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZG
FORT BELVOIR, VIRGINIA 22060

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DEPARTMENT OF THE ARMY
U. S. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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FORT BELVOIR, VIRGINIA 22060

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